Week 6

- A bit more on posets
- Functions
- Countability

Bonus Week 5

1. (1 point) (⋆)
   What does it mean if a relation is antisymmetric?

Solution: A relation ρ on a set A is antisymmetric if

$$a \rho b \wedge b \rho a \implies a = b$$

not $a \neq b$!

is true for all $a, b \in A$. Equivalently, it is antisymmetric if $\rho \cap \hat{\rho} \subseteq \mathrm{id}$.

Consider a set A and a partial order relation ρ. For every $a \in A$, we define

$$\langle a \rangle_\rho := \{b \in A : b \rho^2 a\}.$$

Prove or disprove: The relation σ on A defined as

$$a \sigma b \iff \langle a \rangle_\rho \subseteq \langle b \rangle_\rho$$

is antisymmetric.

Solution:

We prove the claim in the following.

First, we show that ρ^2 is antisymmetric. Consider any $a, b \in A$ such that $a \rho^2 b$. Then there exists some $c \in A$ such that $a \rho c$ and $c \rho b$. By the transitivity of ρ we have $a \rho b$. Hence, for any $a, b \in A$ such that $a \rho^2 b$ and $b \rho^2 a$ we have $a \rho b$ and $b \rho a$. By the antisymmetry of ρ we have $a = b$.

Next we show that σ is antisymmetric. Let $a, b \in A$ be arbitrary and let $a \sigma b$ and $b \sigma a$. Then, we have

$$\langle a \rangle_\rho \subseteq \langle b \rangle_\rho \wedge \langle b \rangle_\rho \subseteq \langle a \rangle_\rho \implies \langle a \rangle_\rho = \langle b \rangle_\rho.$$

As ρ is a partial order relation, it is reflexive. Hence, $a \rho a$ and $b \rho b$ and by definition of ρ^2 we have $a \rho^2 a$ and $b \rho^2 b$. Therefore, $a \in \langle a \rangle_\rho$ and $b \in \langle b \rangle_\rho$. Hence, $a \in \langle b \rangle_\rho$ and $b \in \langle a \rangle_\rho$ as $\langle a \rangle_\rho = \langle b \rangle_\rho$. By the definition of those sets we have $a \rho^2 b$ and $b \rho^2 a$. Thus, by the antisymmetry of ρ^2 we have $a = b$.

Other proofs:

① Show that $\rho^2 = \rho$:

$\rho^2 \subseteq \rho$: Lemma in script

$\rho \subseteq \rho^2$: use reflexivity

$a \rho b \wedge b \rho b \Rightarrow a \rho^2 b$

for all $a, b \in A$, $a \sigma b \wedge b \sigma a$

$\Leftrightarrow \langle a \rangle_\rho \subseteq \langle b \rangle_\rho \wedge \langle b \rangle_\rho \subseteq \langle a \rangle_\rho$

$\Leftrightarrow \langle a \rangle_\rho = \langle b \rangle_\rho$

$b \rho b \Rightarrow b \in \langle b \rangle_\rho = \langle a \rangle_\rho$

$\Rightarrow b \rho a$

$a \rho a \Rightarrow a \in \langle a \rangle_\rho = \langle b \rangle_\rho$

$\Rightarrow a \rho b$

$a \rho b \wedge b \rho a \Rightarrow a = b$

Extra question: What kind of relation is σ?

partial order relation

Definition 3.28. The *direct product* of posets $(A; \preceq)$ and $(B; \sqsubseteq)$, denoted $(A; \preceq) \times (B; \sqsubseteq)$, is the set $A \times B$ with the relation $\leq$ (on $A \times B$) defined by

$$(a_1, b_1) \leq (a_2, b_2) \overset{\text{def}}{\iff} a_1 \preceq a_2 \wedge b_1 \sqsubseteq b_2.$$

Theorem 3.12. *$(A; \preceq) \times (B; \sqsubseteq)$ is a partially ordered set.*

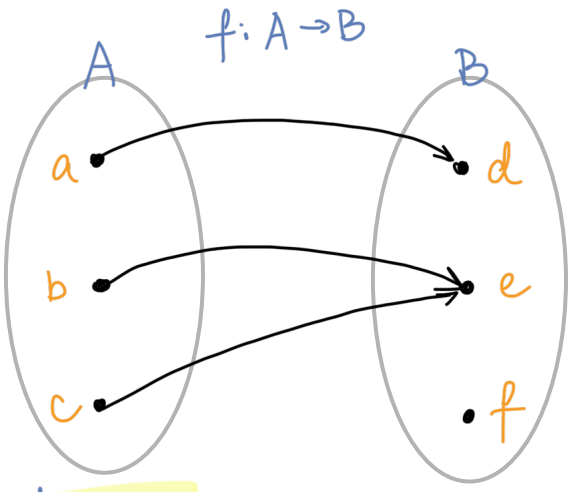
Theorem 3.13. *For given posets $(A; \preceq)$ and $(B; \sqsubseteq)$, the relation $\leq_{\text{lex}}$ defined on $A \times B$ by*

$$(a_1, b_1) \leq_{\text{lex}} (a_2, b_2) \overset{\text{def}}{\iff} a_1 \prec a_2 \vee (a_1 = a_2 \wedge b_1 \sqsubseteq b_2)$$

is a partial order relation.[31]

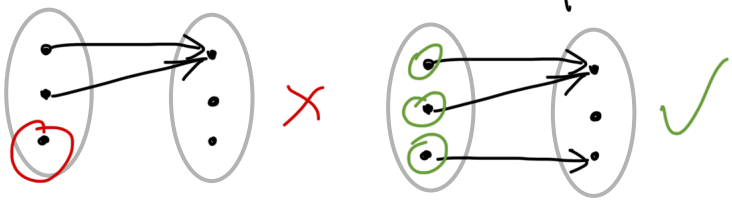
Functions

Terminology:



Functions are relations with special properties:

① totally defined — every input has an output



② well-defined — one input maps to one output

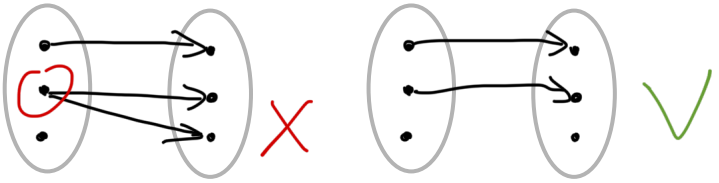


Image $f(S) = \{f(a) \mid a \in S\}$ for $S \subseteq A$

$f(\{a,b,c\}) = \{d,e\}$

$f(\{b,c\}) = \{e\}$

Preimage $f^{-1}(T) = \{a \in A \mid f(a) \in T\}$ for $T \subseteq B$

$f^{-1}(\{e\}) = \{b,c\}$. ≠ inverse function

Composition $g \circ f$ $\quad (g \circ f)(x) = g(f(x))$

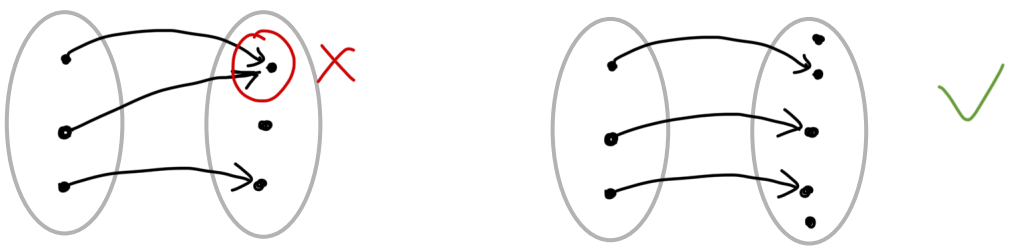
relation:
$x\ f\ y \wedge y\ g\ z$
$x\ (f \circ g)\ z$

$f: A \to B$ is

Injective: $a \neq a' \Rightarrow f(a) \neq f(a')$

equivalently: $f(a) = f(a') \Rightarrow a = a'$

every input is mapped to a different output

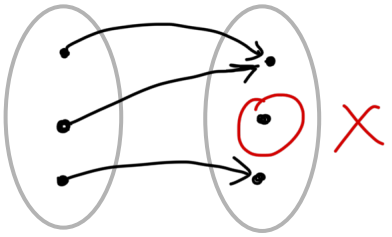


Surjective

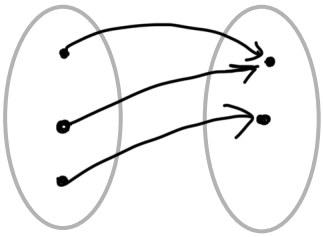
$f(A) = B$

equivallently: $\forall b \in B \ \exists a \in A \ f(a) = b$

every value in the codomain is taken by some input.



X

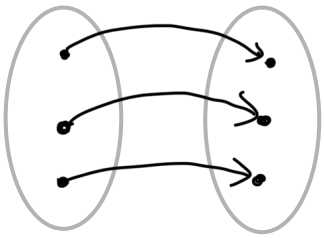


✓

Bijective:

injective and surjective

"one to one correspondence"



$|A| = |B|$

or have the same countability

Inverse function

defined only for bijective functions

$f^{-1}(f(x)) = x \qquad f(f^{-1}(x)) = x$



$x \neq x'$

$\Rightarrow f(x) \neq f(x')$ inj ? surj ?

Example

$f : \mathbb{R}_+ \to \mathbb{R} \quad x \mapsto \sqrt{x}$ ✓ X

$f : \mathbb{R}_+ \to \mathbb{R}_+ \quad x \mapsto \sqrt{x}$ ✓ ✓

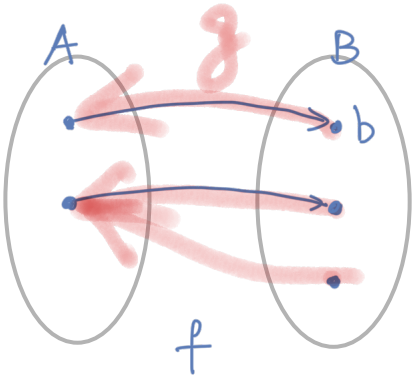
$f : \mathbb{R} \to \mathbb{R} \quad x \mapsto \sqrt{x}$ X totally defined

$f^{-1}(x) = x^2$

FS 16:

d) $(\star\,\star)$ Seien A, B zwei beliebige nicht-leere Mengen. Zeigen Sie folgende Aussage: *(4 Punkte)*

Es existiert $f : A \longrightarrow B$ injektiv. $\iff$ Es existiert $g : B \longrightarrow A$ surjektiv.



totally-defined

trivial

"$\Rightarrow$" We give a surjection $g: B \to A$

Choose a fixed $a_0 \in A$ since $A \neq \emptyset$

$$g(b) = \begin{cases} x \text{ for } x \in f^{-1}(\{b\}) & \text{if } b \in f(A) \\ a_0 & \text{if } b \notin f(A) \end{cases}$$

well-defined

$\forall x, y \in f^{-1}(\{b\})$ $f(x) = f(y) = b$

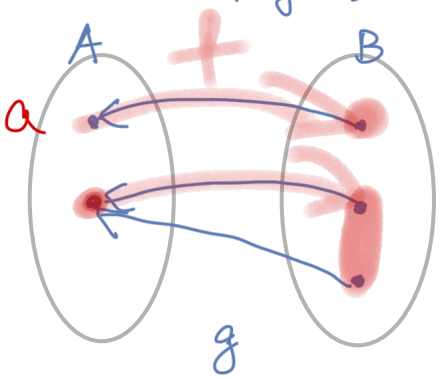
f inj $\Rightarrow x = y$

g: surjective: $\forall a \in A$ $\exists b \in B$ $g(b) = a$

Choose $b = f(a)$ $f(a) \in B$

$g(b) = x$ for $x \in f^{-1}(\{b\}) = a$

"$\Leftarrow$" $\exists\, g: B \to A$ surj. $\Rightarrow$ $\exists\, f: A \to B$ inj



We give an injection $f: A \to B$

$f(a) = b$ for an arbitrary $b \in g^{-1}(\{a\})$

totally defined

$\forall a \in A$ $g^{-1}(\{a\}) \neq \emptyset$

g surj $\forall a$ $\exists b$ $g(b) = a \Rightarrow b \in g^{-1}(\{a\})$

well-defined trivial

f: inj $f(a) = f(a') \Rightarrow g(f(a)) = g(f(a'))$

$\Rightarrow a = a'$

Countability

Definition 3.42.

(i) Two sets A and B *equinumerous*[48], denoted $A \sim B$, if there exists a bijection $A \to B$.

(ii) The set B *dominates* the set A, denoted $A \preceq B$, if $A \sim C$ for some subset $C \subseteq B$ or, equivalently, if there exists an injective function $A \to B$.

(iii) A set A is called *countable*[49] if $A \preceq \mathbb{N}$, and *uncountable*[50] otherwise.[51]

Lemma 3.15. [52]

(i) The relation $\sim$ is an equivalence relation.

(ii) The relation $\preceq$ is transitive: $A \preceq B \wedge B \preceq C \implies A \preceq C$.

(iii) $A \subseteq B \implies A \preceq B$.

Theorem 3.16. $A \preceq B \wedge B \preceq A \implies A \sim B$.

Theorem 3.17. *A set A is countable if and only if it is finite or if $A \sim \mathbb{N}$.*

Proof Patterns:

- prove A countable: Show $A \preceq B$ with B countable

  Then $A \preceq B \preceq \mathbb{N} \Rightarrow A \preceq \mathbb{N}$

- prove A uncountable: Show $B \preceq A$ with B uncountable

  Then if A countable, $B \preceq A \preceq \mathbb{N}$

  $\Rightarrow B \preceq \mathbb{N}$, B countable ↯

- diagonalization argument (rare, but see HS19 1.(e))

Choices for B:

countable	$\mathbb{N}$	$\gtrsim$	$\mathbb{Q}$	$\{0,1\}^*$	
uncountable	$\{0,1\}^\infty$	$\{0,1,2\}^\infty$	$\mathbb{R}$	$\mathcal{P}(\mathbb{N})$	

0 1 0 1 1 0 . . .

1 1 3 4 . .

FS 18

$h: A \cup B \to \mathbb{N}$

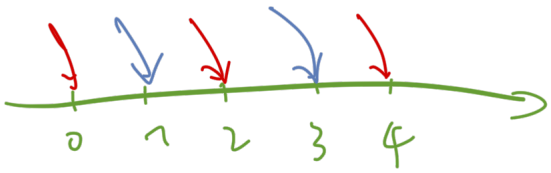
c) $(\star\star)$

1.) Let A and B be two disjoint countable sets. Prove that $A \cup B$ is countable by showing an appropriate injection. *(3 Points)*

$A \preceq \mathbb{N} \wedge B \preceq \mathbb{N}$ $\quad 2^{f(x)+1}$ $\quad 3^{g(x)+1}$

$\exists f, g$ $\quad f: A \to \mathbb{N}$, $g: B \to \mathbb{N}$, f, g inj.

$$h(x) = \begin{cases} 2f(x) & x \in A \\ 2g(x)+1 & x \in B \end{cases}$$



h inj $\quad h(x) = h(x')$ Either $2f(x) = 2f(x')$ $(f, g$ inj$)$ $\Rightarrow x = x'$

or $2g(x)+1 = 2g(x')+1$

Extra question: why disjoint? h well defined!

2.) Using the above fact, prove that if a set A is countable and a set C is uncountable, then $C \setminus A$ is uncountable. *(2 Points)*

Note: You will receive points even without solving 1.)

Assume $C \setminus A$ is countable

$(C \setminus A) \cup A = C \Rightarrow C$ countable ↯

$(C \setminus A)$ countable, A countable

Corollary 3.20. *The Cartesian product $A \times B$ of two countable sets A and B is countable, i.e., $A \preceq \mathbb{N} \wedge B \preceq \mathbb{N} \implies A \times B \preceq \mathbb{N}$.*

$\mathbb{N} \times \mathbb{N}$

Theorem 3.22. *Let A and A_i for $i \in \mathbb{N}$ be countable sets.*

$q = \frac{m}{n}$ $m, n \in \mathbb{Z}$

(i) *For any $n \in \mathbb{N}$, the set A^n of n-tuples over A is countable.*

(ii) *The union $\bigcup_{i \in \mathbb{N}} A_i$ of a countable list $A_0, A_1, A_2, \ldots$ of countable sets is countable.*

(m, n)

(iii) *The set A^* of finite sequences of elements from A is countable.*

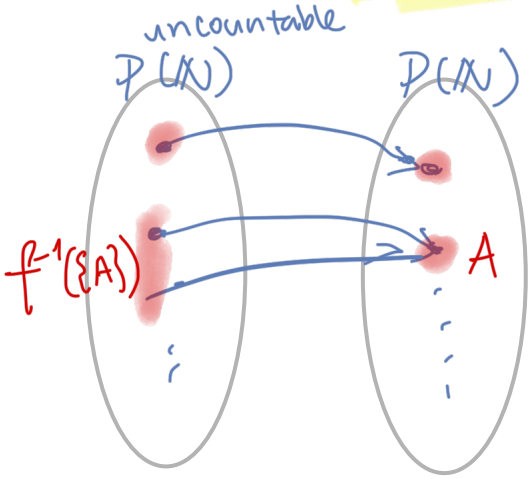
$\mathbb{Q} \subseteq \mathbb{Z} \times \mathbb{Z}$

$\{0,1\}^*$

HS 19

f) $(\star\star)$ Let $f : \mathcal{P}(\mathbb{N}) \to \mathcal{P}(\mathbb{N})$ be any function. Prove that either the image of f, $Im(f)$, is uncountable or there exists an $A \in \mathcal{P}(\mathbb{N})$ such that $f^{-1}(\{A\})$ is uncountable (or both). *(8 Points)*

① ②



Case distinction

① uncountable trivial

① countable, prove ② uncountable

Im(f) countable

Assume $\forall A \in \mathcal{P}(\mathbb{N})$ $f^{-1}(\{A\})$ countable

$$\bigcup_{A \in Im(f)} f^{-1}(\{A\}) = \mathcal{P}(\mathbb{N}) \Rightarrow \mathcal{P}(\mathbb{N}) \text{ countable } \lightning$$

(countable) (countable) (countable)

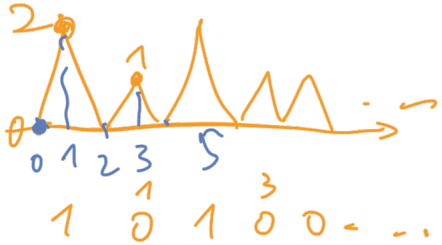
Prove $\mathcal{P}(\mathbb{N})$ uncountable separately!

More on constructing injections

Injections from $\{0,1\}^\infty \to$? to prove uncountable

f) $(\star\star)$ We call a function $f : \mathbb{N} \to \mathbb{N}$ a zigzag function if for all $n > 0$, we have: if $f(n-1) < f(n)$ then $f(n) > f(n+1)$, and if $f(n-1) > f(n)$ then $f(n) < f(n+1)$ (that is, the function alternates between bigger and smaller values). Is the set of all zigzag functions countable? Prove your answer. *(4 Points)*

$\{f : f \text{ is zigzag function}\}$ HS18



$g\ \{0,1\}^\infty \to \{\text{zigzag}\}$

$$g(b) = f_b(x) \begin{cases} 0 & x = 2i \\ 1 & x = 2i+1 \wedge b_i = 0 \\ 2 & x = 2i+1 \wedge b_i = 1 \end{cases}$$

Show g inj.

$b_i = \begin{cases} 0 \\ 1 \end{cases}$ b_i : ith bit

$x = 2i+1$

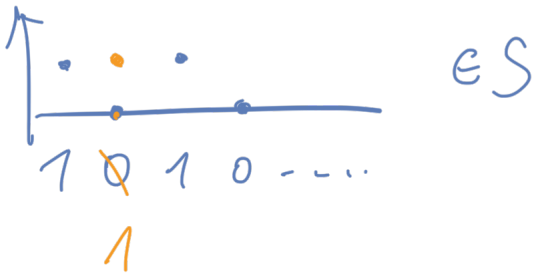
$b \neq b' \Rightarrow \exists i\ b_i \neq b_i'$

$\Rightarrow g(b) \neq g(b')$

because $g(b)(2i+1) \neq g(b')(2i+1)$

Exercise: prove $S = \{f : \mathbb{N} \to \mathbb{N}\}$ is uncountable

$g : \{0,1\}^\infty \to S$ Find inj g



Summary for proofs in countability

- prove A countable: Show $A \preceq B$ with B countable
- prove A uncountable: Show $B \preceq A$ with B uncountable

↑ with injections !

① use given injections (countable sets)
② $\{0,1\}^\infty$ draw some functions
③ prime numbers ④ ... ⑤ ...

- use theorems for more complex sets

Corollary 3.20. *The Cartesian product $A \times B$ of two countable sets A and B is countable, i.e., $A \preceq \mathbb{N} \wedge B \preceq \mathbb{N} \implies A \times B \preceq \mathbb{N}$.*

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